SMOKING BEHAVIOUR AND DEMOGRAPHIC ANALYSIS

SQL Phase 2 Project Report

A reproducible SQLite investigation of smoking prevalence, demographic variation and consumption intensity

Project element	Details
Dataset	Smoking survey - 1,691 respondents
Database engine	SQLite
Main table	smoking_survey
SQL portfolio	75 validated queries
Difficulty structure	25 easy, 25 medium, 25 difficult
Report status	Detailed audited submission version
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Prepared for academic assessment and viva discussion

Executive summary

This project builds and analyses a relational SQLite database from a smoking survey containing 1,691 respondents. The analysis identifies 421 current smokers and 1,270 non-smokers, giving a sample smoking prevalence of 24.90%. The work is designed not only to answer descriptive questions but also to demonstrate a complete SQL workflow: data preparation, schema creation, validation, basic retrieval, grouped analysis, conditional aggregation, common table expressions, subqueries, reusable views and window functions.

The strongest descriptive patterns in the sample concern age, marital status, qualification and region. Smoking prevalence is highest among respondents aged 18-24 (37.24%) and 25-34 (35.11%), and lowest among those aged 65 or over (11.41%). Single and divorced respondents have rates of 37.00% and 36.02%, compared with 17.61% among married respondents. Scotland has the highest regional sample rate at 34.46%, while Midlands & East Anglia has the lowest at 20.77%. These are sample associations, not causal findings.

Among current smokers, estimated consumption averages 101.57 cigarettes per week. Weekend consumption averages 16.41 cigarettes per day compared with 13.75 on weekdays. Most smokers either report no weekend change (230 people) or an increase (166 people), while 25 report a decrease. Packet cigarettes are the most common product type, although the smaller mixed-product and hand-rolled groups show higher average weekly intensity.

Key audit result
All 75 query files execute successfully against the packaged database. The rebuilt database contains 1,691 unique respondent IDs, exactly two smoking-status values, no smokers with incomplete smoker-specific details, and no non-smokers with unexpected smoking details.

1. Introduction and project context

1.1 Background

Smoking behaviour can be examined from several connected perspectives: whether a respondent smokes, how much a smoker consumes, whether behaviour differs between weekdays and weekends, and how these outcomes vary across demographic groups. SQL is well suited to this type of investigation because it supports reproducible filtering, aggregation, ranking and validation while retaining a clear audit trail.

1.2 Aim

The aim is to design a reliable SQLite project that transforms the supplied survey file into a documented analytical database and answers 75 progressively more demanding questions about smoking behaviour and respondent characteristics.

1.3 Objectives

- Create a clean SQLite table with stable column names and a unique respondent identifier.
- Preserve genuine missing values and distinguish not-applicable fields from zero consumption.
- Measure smoking prevalence overall and across demographic dimensions.
- Analyse smoker intensity, product type and weekday/weekend changes.
- Demonstrate easy, medium and difficult SQL techniques in a structured sequence.
- Create reusable views so repeated calculations are defined consistently.
- Validate every query and provide evidence suitable for assessment and viva discussion.

1.4 Research questions

- What proportion of the survey respondents currently smoke?
- Which demographic groups have higher or lower sample smoking prevalence?
- How does estimated weekly consumption vary across smoker groups?
- How does weekend consumption compare with weekday consumption?
- Which SQL techniques provide the clearest and most reusable solution to each analytical problem?
- Does the final database pass structural and logical integrity tests?

2. Dataset description

2.1 Source structure

The supplied source contains 1,691 observations and 12 original survey variables. The project database adds five derived fields, producing a 17-column analytical table. Every row represents one respondent; the dataset is therefore at respondent level rather than transaction level.

Column	Type	Purpose
respondent_id	INTEGER	Project-generated sequential identifier; unique row key.
gender	TEXT	Respondent gender category.
age	INTEGER	Age in completed years.
marital_status	TEXT	Recorded marital status.
highest_qualification	TEXT	Highest educational qualification category.
nationality	TEXT	Recorded nationality.
ethnicity	TEXT	Recorded ethnicity.

Column	Type	Purpose
gross_income	TEXT	Categorical annual gross-income band.
region	TEXT	UK regional category.
smoke	TEXT	Current smoking status: Yes or No.
amt_weekends	REAL	Reported cigarettes per weekend day for smokers.
amt_weekdays	REAL	Reported cigarettes per weekday for smokers.
type	TEXT	Smoking product type for smokers.
avg_daily_cigarettes	REAL	Derived mean of weekday and weekend daily amounts.
estimated_weekly_cigarettes	REAL	Derived as 5 weekday days plus 2 weekend days.
age_group	TEXT	Derived ordered analytical age band.
income_band_order	INTEGER	Derived numeric sort key for income bands.

2.2 Null-value interpretation

There are 1,270 null values in each smoker-specific source field: weekday amount, weekend amount and smoking type. These nulls correspond exactly to the 1,270 non-smokers. They are retained as NULL because the questions are not applicable to those respondents. Replacing these values with zero would incorrectly imply that a non-smoker supplied a measured consumption value and would distort averages if the smoker filter were omitted.

Null-safe analytical rule

Any calculation of smoking amount or product type must filter WHERE smoke = 'Yes' or use an equivalent conditional expression. NULL means not applicable in this dataset, not missing evidence of zero cigarettes.

3. Data preparation and database design

3.1 Cleaning decisions

- Trimmed and standardised text values without recoding substantive responses.
- Converted age and cigarette-amount fields to numeric types.
- Retained category wording to preserve traceability to the source file.
- Generated respondent_id values from 1 to 1,691 for stable identification.
- Created age_group and income_band_order fields to support meaningful ordered output.
- Kept smoker-only fields NULL for non-smokers.

3.2 Derived measures

Two consumption measures were derived for smokers. The average-daily field provides a simple midpoint between weekday and weekend daily amounts, while the estimated-weekly field weights weekdays and weekends according to the number of days in a normal week.

```
avg_daily_cigarettes = (amt_weekdays + amt_weekends) / 2.0
estimated_weekly_cigarettes = (amt_weekdays * 5) + (amt_weekends * 2)
```

The weekly estimate is used for most intensity comparisons because it reflects the different number of weekday and weekend days. It remains an estimate based on self-reported typical daily amounts, rather than a directly observed weekly total.

3.3 Table and view strategy

A single respondent-level table is appropriate because the source is a compact denormalised survey extract and contains no repeating child records. Five views were created to centralise recurring calculations and avoid rewriting group logic. The views support regional prevalence, age-group prevalence, income summaries, smoker details and general category analysis.

Views improve consistency because the same numerator, denominator, percentage formula and ordering can be reused by later queries. They also make difficult queries easier to read during a viva: the analytical intention is visible without repeating every implementation detail.

4. SQL query portfolio and progression

4.1 Easy queries: Q01-Q25

The first 25 queries establish familiarity with the table and confirm that simple retrieval is correct before advanced calculations are attempted. They use SELECT, COUNT, DISTINCT, WHERE, BETWEEN, ORDER BY, LIMIT, GROUP BY, basic averages, NULL counts and CASE expressions. This stage also checks the central business rule that smoker-specific details align with smoking status.

- Q01-Q04 inspect the data, row count, categories and age range.
- Q05-Q15 retrieve smokers and non-smokers and summarise demographic and product categories.
- Q16-Q23 introduce age bands and basic smoking-intensity averages.

- Q24-Q25 test missing-value patterns and status/detail consistency.

4.2 Medium queries: Q26-Q50

The middle 25 queries turn raw counts into comparative measures. Conditional aggregation calculates smoking rates using a consistent formula: smokers divided by all respondents in the same group. HAVING is used to protect selected comparisons from extremely small group sizes. CASE expressions classify weekend change and consumption bands, while cross-tabulations display smokers and non-smokers in the same result.

```
ROUND(100.0 * SUM(CASE WHEN smoke = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS smoking_rate_pct
```

- Q26-Q34 compare overall and group-specific prevalence.
- Q35-Q41 analyse weekly intensity, weekday/weekend behaviour and consumption bands.
- Q42-Q45 identify high daily consumption and compare counts and average ages.
- Q46-Q50 perform further quality checks and intensity comparisons by income and qualification.

4.3 Difficult queries: Q51-Q75

The final 25 queries demonstrate advanced analytical SQL. CTEs make multi-stage calculations readable, window functions preserve row-level or group-level detail while adding ranks and shares, and correlated subqueries compare each smoker with a relevant local benchmark. UNION ALL combines labelled extremes, and reusable views are queried directly near the end of the portfolio.

- DENSE_RANK orders regions and age groups by smoking rate.
- Window percentages calculate each gender or region's share of all smokers.
- ROW_NUMBER identifies top smokers within gender and region.
- NTILE assigns smokers to consumption quartiles.
- LAG measures changes across ordered age and income categories.
- Correlated and joined benchmarks compare individuals with region, gender or age-group averages.
- Q75 consolidates row, status and detail checks into one PASS/FAIL validation result.

5. Descriptive results

5.1 Overall profile

Measure	Result
Respondents	1,691
Current smokers	421
Non-smokers	1,270

Measure	Result
Overall smoking rate	24.90%
Average respondent age	49.84 years
Age range	16-97 years

Approximately one quarter of the sample identifies as a current smoker. The overall rate is the baseline used in later difference calculations, especially Q61, where each regional rate is measured against 24.90 percentage points.

5.2 Regional variation

Region	Respondents	Smokers	Rate
Scotland	148	51	34.46%
London	182	50	27.47%
South West	157	42	26.75%
South East	252	65	25.79%
Wales	83	20	24.10%
The North	426	101	23.71%
Midlands & East Anglia	443	92	20.77%

Scotland has the highest observed regional smoking rate at 34.46%, 13.69 percentage points above Midlands & East Anglia at 20.77%. London, South West and South East fall above the overall sample rate, while Wales, The North and Midlands & East Anglia fall below it. Regional results should not be interpreted as direct regional effects because age, income, education and other group composition may differ.

5.3 Gender comparison

Gender	Respondents	Smokers	Rate
Male	726	187	25.76%

Gender	Respondents	Smokers	Rate
Female	965	234	24.25%

The male sample rate is 25.76% and the female rate is 24.25%, a difference of 1.51 percentage points. Although women account for more smokers in absolute terms (234 versus 187), they also form a larger share of the total sample. This demonstrates why prevalence rates and raw counts answer different questions.

5.4 Age-group comparison

Age group	Respondents	Smokers	Rate
18-24	145	54	37.24%
25-34	282	99	35.11%
35-44	325	95	29.23%
45-54	250	75	30.00%
55-64	242	47	19.42%
65+	447	51	11.41%

Smoking prevalence is highest in the two youngest analytical groups: 37.24% for ages 18-24 and 35.11% for ages 25-34. Rates remain around 29-30% for ages 35-54, then fall to 19.42% for ages 55-64 and 11.41% for ages 65+. The pattern is descriptive and may reflect cohort, survivorship, cessation and sampling effects.

5.5 Marital-status comparison

Marital status	Respondents	Smokers	Rate
Single	427	158	37.00%
Divorced	161	58	36.02%
Separated	68	22	32.35%

Marital status	Respondents	Smokers	Rate
Widowed	223	40	17.94%
Married	812	143	17.61%

Single, divorced and separated respondents have the highest observed smoking rates. Married and widowed groups have much lower rates. Age is a likely confounding factor because marital-status categories have different age distributions, so the result should be presented as an association within this sample rather than as evidence that marital status causes smoking.

5.6 Qualification comparison

Highest qualification	Respondents	Smokers	Rate
GCSE/CSE	102	38	37.25%
GCSE/O Level	308	105	34.09%
ONC/BTEC	76	23	30.26%
Other/Sub Degree	127	31	24.41%
No Qualification	586	137	23.38%
Higher/Sub Degree	125	27	21.60%
A Levels	105	21	20.00%
Degree	262	39	14.89%

The highest observed rates occur in the GCSE/CSE and GCSE/O Level groups, while degree holders have the lowest rate at 14.89%. The no-qualification group is the largest category but does not have the highest rate. This is another example of why group size and within-group percentage must be considered together.

5.7 Income-band comparison

Income band	Respondents	Smokers	Rate
Under 2,600	133	36	27.07%
2,600 to 5,200	257	64	24.90%
5,200 to 10,400	396	107	27.02%
10,400 to 15,600	268	83	30.97%
15,600 to 20,800	188	45	23.94%
20,800 to 28,600	155	38	24.52%
28,600 to 36,400	79	9	11.39%
Above 36,400	89	15	16.85%
Refused	108	21	19.44%
Unknown	18	3	16.67%

The income pattern is not perfectly linear. The 10,400 to 15,600 band has the highest rate at 30.97%, while the two highest specified bands have lower rates. 'Refused' and 'Unknown' are kept as separate categories rather than forced into the income ordering because they do not represent measured income levels.

6. Smoking-intensity results

6.1 Weekday, weekend and weekly estimates

Intensity measure	Value
Average weekday amount	13.75 cigarettes/day
Average weekend amount	16.41 cigarettes/day
Average estimated weekly amount	101.57 cigarettes/week
Minimum estimated weekly amount	0

Intensity measure	Value
Maximum estimated weekly amount	350

Average reported consumption is 2.66 cigarettes per day higher at weekends than on weekdays. The weekly estimate varies widely, from 0 to 350 cigarettes, showing substantial heterogeneity among current smokers. A recorded value of zero is retained because it is present for a respondent classified as a current smoker; it should be reviewed rather than automatically removed.

6.2 Weekend behaviour

Pattern	Smokers	Share of smokers
No change	230	54.63%
Increase	166	39.43%
Decrease	25	5.94%

More than half of smokers report the same amount on weekdays and weekends. About 39.43% report an increase at weekends, and 5.94% report a decrease. Q39 calculates the mean numerical change, while Q40 classifies the direction, demonstrating that averages and category counts provide complementary interpretations.

6.3 Consumption bands

Estimated weekly band	Smokers	Share
Under 35	51	12.11%
35-69	79	18.76%
70-139	159	37.77%
140+	132	31.35%

The two highest bands contain 291 smokers, or 69.12% of current smokers. These bands are analytical categories created for this project and should not be treated as clinical classifications. Their purpose is to demonstrate CASE-based segmentation and make the distribution easier to discuss.

6.4 Product type

Smoking type	Smokers	Average weekly estimate
Packets	297	96.56
Hand-Rolled	72	111.40
Both/Mainly Packets	42	115.90
Both/Mainly Hand-Rolled	10	119.50

Packets are used by 297 smokers and therefore dominate the product mix. The smaller groups involving hand-rolled products show higher average weekly estimates, but their sample sizes are much smaller, especially the 'Both/Mainly Hand-Rolled' group with only 10 smokers. The report therefore avoids presenting the highest average as automatically the most reliable estimate.

7. Advanced analytical interpretation

7.1 Ranking and contribution

Q51 and Q52 rank regional and age-group prevalence using window functions. Unlike ORDER BY alone, DENSE_RANK explicitly assigns a rank that remains available as a result column. Q53 and Q54 calculate each gender's and region's contribution to the total smoker count, separating composition from prevalence. For example, The North contributes the largest number of smokers because it has a large respondent base, even though its within-region rate is below the sample average.

7.2 Individual versus benchmark comparisons

Q58 compares every smoker with the overall smoker mean. Q59 uses a correlated subquery to compare each smoker with the mean for that smoker's own region. Q60 creates gender averages in a CTE and joins them back to smoker records. Q72 performs an age-group comparison using a window average. These queries answer increasingly contextual questions and demonstrate several valid SQL approaches to benchmark analysis.

7.3 Ordered trend analysis

Q62 and Q63 use LAG after constructing ordered group summaries. This is important because text labels do not always sort in analytical order. Age groups are ordered using their minimum age, and income groups use the project-generated income_band_order. The LAG result shows the change from the immediately preceding category rather than from the overall average.

7.4 Reusable views

Q73 and Q74 query pre-built views rather than repeating the full aggregation. This demonstrates abstraction: a view can become a stable analytical interface used by reports, dashboards or later SQL files. The principal advantage is consistency; if the underlying formula needs correction, it can be changed once in the view definition.

8. Data quality, validation and audit evidence

Validation test	Result	Interpretation
Total rows	1,691	Matches cleaned source and rebuilt database.
Unique respondent IDs	1,691	No duplicate project identifiers.
Distinct smoking statuses	2	Only Yes and No are present.
Smokers missing details	0	No incomplete smoker-specific records.
Non-smokers with details	0	No contradictory smoker-specific values.
Query execution	75/75 PASS	Every individual SQL file executed successfully.

8.1 Query-level execution evidence

The validation CSV records a PASS status, output row count, output column count and error field for every query. Different row counts are expected because queries answer different questions. A one-row output may be an aggregate; a 421-row output may assign a calculation to every smoker; a limited result intentionally returns only the requested top records.

8.2 Rebuild test

The full project SQL file was executed to rebuild the table and views independently. The rebuilt table produced the same 1,691-row count and passed the final validation query. This verifies that the package is reproducible and not dependent solely on the pre-built SQLite file.

8.3 Logical controls

- Smoker-specific averages always exclude non-smokers.
- Percentage denominators are calculated within the relevant group.
- Income and age labels are ordered using explicit numeric logic.

- Minimum subgroup thresholds are used in selected nationality, ethnicity and qualification comparisons.
- Queries use ROUND for readable percentage and average outputs.
- Advanced calculations preserve transparency through named CTEs and views.

9. Limitations and ethical interpretation

9.1 Analytical limitations

- The data are cross-sectional, so the analysis cannot establish cause and effect.
- Smoking and consumption amounts are self-reported and may be affected by recall or social-desirability bias.
- The project does not contain survey weights, so results should not automatically be generalised to the wider population.
- Some nationality, ethnicity and smoking-type groups are small, increasing uncertainty in their percentages or averages.
- Income is categorical rather than continuous, and refused/unknown responses cannot be ordered as measured income.
- The estimated weekly measure assumes reported weekday and weekend amounts are typical.
- The dataset contains no time dimension, preventing analysis of change over years.

9.2 Ethical communication

Demographic comparisons should be communicated without stereotyping. A higher observed rate in a category does not imply that every member of that group behaves similarly, nor does it identify the cause of the difference. Results should be framed as sample-level associations, accompanied by group sizes and suitable caveats.

Responsible conclusion
The SQL outputs describe this supplied sample. They do not provide medical advice, clinical risk classification or proof that demographic characteristics cause smoking behaviour.

10. Conclusion and recommendations

The project successfully converts the supplied survey into a reproducible SQLite analysis with 75 validated queries. It demonstrates a clear progression from basic data retrieval to advanced ranking, benchmarking, trend and view-based analysis. The most prominent sample findings are the age gradient in smoking prevalence, high rates among single and divorced respondents, meaningful regional variation and higher average consumption at weekends.

From a technical perspective, the project shows that reliable SQL analysis depends on more than producing a result. Correct null handling, consistent denominators, explicit ordering, subgroup-size awareness, reusable views and independent validation are all necessary for a defensible submission.

10.1 Recommendations for future analysis

- Add survey weights if available and compare weighted with unweighted estimates.
- Use multivariable statistical modelling outside SQLite to examine confounding among age, marital status, education, income and region.
- Collect or integrate time-series information to examine smoking trends and cessation.
- Investigate the smoker with a zero weekly estimate and document whether it represents occasional smoking or a response anomaly.
- Create visual dashboards from the validated views while retaining the SQL database as the source of truth.

Appendix A - Complete 75-query catalogue

The table below maps every SQL file to its analytical perspective, difficulty, principal concept and business question. The execution-status column comes from the package validation evidence.

Query	Level	SQL concept	Question answered	Audit
Q01	Easy	SELECT	What are the first 10 survey records?	PASS (10 rows)
Q02	Easy	COUNT	How many respondents are in the survey?	PASS (1 rows)
Q03	Easy	DISTINCT	Which gender values occur in the dataset?	PASS (2 rows)
Q04	Easy	MIN/MAX/AVG	What are the minimum, maximum and average ages?	PASS (1 rows)
Q05	Easy	WHERE	Which respondents currently smoke?	PASS (50 rows)
Q06	Easy	WHERE	Which respondents do not smoke?	PASS (50 rows)
Q07	Easy	GROUP BY	How many smokers and non-smokers are there?	PASS (2 rows)
Q08	Easy	GROUP BY	How many respondents are in each region?	PASS (7 rows)
Q09	Easy	GROUP BY	How many respondents are in each marital-status group?	PASS (5 rows)
Q10	Easy	GROUP BY	How many respondents have each qualification?	PASS (8 rows)
Q11	Easy	GROUP BY	How many respondents are in each income band?	PASS (10 rows)
Q12	Easy	IS NOT NULL	Which smokers have recorded weekday amounts?	PASS (50 rows)
Q13	Easy	ORDER BY	Which smokers report the highest weekend amount?	PASS (10 rows)
Q14	Easy	DISTINCT	Which smoking types are	PASS (4 rows)

Query	Level	SQL concept	Question answered	Audit
			recorded?	
Q15	Easy	GROUP BY	How many smokers use each recorded type?	PASS (4 rows)
Q16	Easy	BETWEEN	Which respondents are aged 25 to 34?	PASS (100 rows)
Q17	Easy	GROUP BY	How many respondents are in each age group?	PASS (6 rows)
Q18	Easy	GROUP BY	What are the most common nationalities?	PASS (8 rows)
Q19	Easy	GROUP BY	What are the ethnicity counts?	PASS (7 rows)
Q20	Easy	WHERE/GROUP BY	How many smokers are in each region?	PASS (7 rows)
Q21	Easy	WHERE/GROUP BY	How many smokers are in each gender group?	PASS (2 rows)
Q22	Easy	AVG	What is the average weekday amount among smokers?	PASS (1 rows)
Q23	Easy	AVG	What is the average weekend amount among smokers?	PASS (1 rows)
Q24	Easy	NULL COUNT	How many rows have missing smoking-amount values?	PASS (1 rows)
Q25	Easy	CASE	Are smoker-specific fields consistent with smoking status?	PASS (2 rows)
Q26	Medium	Conditional Aggregation	What is the overall smoking rate?	PASS (1 rows)
Q27	Medium	Conditional Aggregation	What is the smoking rate in each region?	PASS (7 rows)
Q28	Medium	Conditional Aggregation	What is the smoking rate by gender?	PASS (2 rows)

Query	Level	SQL concept	Question answered	Audit
Q29	Medium	Conditional Aggregation	What is the smoking rate by age group?	PASS (6 rows)
Q30	Medium	Conditional Aggregation	What is the smoking rate by qualification?	PASS (8 rows)
Q31	Medium	Conditional Aggregation	What is the smoking rate by income band?	PASS (10 rows)
Q32	Medium	Conditional Aggregation	What is the smoking rate by marital status?	PASS (5 rows)
Q33	Medium	HAVING	Which nationalities with at least 20 respondents have the highest smoking rates?	PASS (6 rows)
Q34	Medium	HAVING	Which ethnicity groups with at least 20 respondents have the highest smoking rates?	PASS (4 rows)
Q35	Medium	GROUP BY/AVG	What is average weekly cigarette consumption by gender among smokers?	PASS (2 rows)
Q36	Medium	GROUP BY/AVG	What is average weekly consumption by age group among smokers?	PASS (6 rows)
Q37	Medium	GROUP BY/AVG	What is average weekly consumption by region among smokers?	PASS (7 rows)
Q38	Medium	GROUP BY/AVG	How does average consumption vary by smoking type?	PASS (4 rows)
Q39	Medium	Expression	What is the average weekday-to-weekend change among smokers?	PASS (1 rows)
Q40	Medium	CASE/GROUP BY	How many smokers increase, decrease or maintain consumption at weekends?	PASS (3 rows)
Q41	Medium	CASE	How many smokers fall into weekly consumption bands?	PASS (4 rows)

Query	Level	SQL concept	Question answered	Audit
Q42	Medium	WHERE	Which smokers report at least 20 cigarettes on an average day?	PASS (143 rows)
Q43	Medium	Conditional Aggregation	How does smoking status vary by gender in counts?	PASS (2 rows)
Q44	Medium	Conditional Aggregation	How does smoking status vary by region in counts?	PASS (7 rows)
Q45	Medium	GROUP BY/AVG	What is the average age of smokers and non-smokers?	PASS (2 rows)
Q46	Medium	CASE	Are there any smokers without complete smoking details?	PASS (1 rows)
Q47	Medium	CASE	Are there non-smokers with unexpected smoking details?	PASS (1 rows)
Q48	Medium	GROUP BY	Which regions have the youngest and oldest average respondents?	PASS (7 rows)
Q49	Medium	GROUP BY/AVG	What is smoker intensity by income band?	PASS (10 rows)
Q50	Medium	HAVING	Which qualification groups with at least 10 smokers have the highest intensity?	PASS (8 rows)
Q51	Difficult	CTE/Window	How do regions rank by smoking rate?	PASS (7 rows)
Q52	Difficult	CTE/Window	How do age groups rank by smoking rate?	PASS (6 rows)
Q53	Difficult	Window Function	What share of all smokers belongs to each gender?	PASS (2 rows)
Q54	Difficult	Window Function	What share of all smokers comes from each region?	PASS (7 rows)
Q55	Difficult	Window Function	Who are the top five smokers by weekly consumption within each	PASS (10 rows)

Query	Level	SQL concept	Question answered	Audit
			gender?	
Q56	Difficult	Window Function	Who are the top three smokers by weekly consumption within each region?	PASS (21 rows)
Q57	Difficult	NTILE	Which weekly consumption quartile is each smoker in?	PASS (421 rows)
Q58	Difficult	Subquery	Which smokers consume more than the smoker average?	PASS (200 rows)
Q59	Difficult	Correlated Subquery	Which smokers consume more than their own region average?	PASS (184 rows)
Q60	Difficult	CTE/JOIN	Which smokers consume more than their gender average?	PASS (185 rows)
Q61	Difficult	CTE	How far is each regional smoking rate from the overall rate?	PASS (7 rows)
Q62	Difficult	LAG	How does smoking rate change between adjacent age groups?	PASS (6 rows)
Q63	Difficult	LAG	How does smoking rate change across ordered income bands?	PASS (10 rows)
Q64	Difficult	Window Function	What is the cumulative share of smokers by region?	PASS (7 rows)
Q65	Difficult	Conditional Aggregation	What is the smoking-type mix within each gender?	PASS (2 rows)
Q66	Difficult	Conditional Aggregation	What is the smoking-type mix within each region?	PASS (7 rows)
Q67	Difficult	Window Function	What is each smoker consumption value compared with the previous ranked smoker?	PASS (50 rows)

Query	Level	SQL concept	Question answered	Audit
Q68	Difficult	CTE/UNION	Which regions have the highest and lowest smoking rates?	PASS (2 rows)
Q69	Difficult	CTE/UNION	Which smoker groups have the highest and lowest average weekly consumption?	PASS (2 rows)
Q70	Difficult	GROUP BY/HAVING	Which gender-age groups with at least 20 respondents have the highest smoking rates?	PASS (12 rows)
Q71	Difficult	GROUP BY/HAVING	Which region-gender groups with at least 20 respondents have the highest smoking rates?	PASS (14 rows)
Q72	Difficult	Window Function	How does each smoker differ from the average consumption of their age group?	PASS (421 rows)
Q73	Difficult	VIEW	What are the smoking-rate results from the reusable regional view?	PASS (7 rows)
Q74	Difficult	VIEW	What are the ordered income smoking summaries from the reusable view?	PASS (10 rows)
Q75	Difficult	CTE/CASE	Does the table pass core row, status and smoker-detail checks?	PASS (1 rows)

Appendix B - Selected SQL patterns explained

Conditional aggregation

Used when a grouped result requires both a total denominator and a conditional numerator. SUM(CASE WHEN smoke = 'Yes' THEN 1 ELSE 0 END) counts smokers without removing non-smokers from the denominator.

HAVING

Filters completed groups rather than individual rows. It is used to require a minimum number of respondents or smokers before a rate or average is compared.

Common table expression

A WITH clause names an intermediate result. This improves readability when a group summary must be ranked, joined or compared with an overall value.

Correlated subquery

A subquery refers to a value from the current outer row. Q59 uses this to calculate the relevant regional average for each smoker.

Window function

Calculates ranks, shares, previous values or cumulative totals without collapsing all rows into one row per group.

View

Stores a reusable SELECT definition. A view does not duplicate the underlying survey data; it exposes a consistent query interface.

UNION ALL

Stacks labelled result sets and keeps all rows. It is used for highest/lowest comparisons where both extreme records must appear in one output.

NULL handling

IS NULL and IS NOT NULL are required because equality comparisons do not treat NULL as an ordinary value. Smoker-only measures are filtered explicitly.

Appendix C - Viva-ready defence points

Why was SQLite selected?

It is portable, requires no server configuration, supports the SQL features used in the project and allows the database to be submitted as a single file.

Why keep NULL values for non-smokers?

The smoker-specific questions are not applicable. Zero is a measured quantity, while NULL records the absence of an applicable value.

Why use estimated weekly consumption?

It applies five weekday days and two weekend days, so it is more representative than a simple average when comparing weekly intensity.

Why are rates preferable to counts for group comparison?

Counts are influenced by group size. Rates divide smokers by all respondents in the same group and therefore support fairer comparison.

Why are some groups filtered with HAVING?

Very small denominators can produce unstable percentages or averages. A minimum group size improves interpretability.

What does a window function add?

It adds rankings, shares or benchmarks while retaining each result row, avoiding unnecessary collapse of detail.

Can the results prove causation?

No. The data are observational and cross-sectional, and the SQL analysis is descriptive.

How was correctness checked?

Every query was executed, row and column counts were logged, the complete SQL build was independently rebuilt, and logical data-quality checks were consolidated in Q75.

Appendix D - Submission checklist

Check	Required action
Student identity	Cover name updated to Shraddha Yadav.
Database	Open the packaged .sqlite file and run the first test query.
SQL file	Confirm the full project SQL opens as plain text.
Report	Review the Word/PDF report and insert any required screenshots.
Evidence	Keep QUERY_OUTPUT_VALIDATION_75.csv with the

Check	Required action
	working package.
Approval	Submit the final-submission ZIP only after dataset approval requirements are satisfied.
Integrity	Do not rename the main table unless all 75 queries and views are updated consistently.